

The botanical wallpaper in Carl Linnaeus's former home, now a museum, in Uppsala, Sweden, is a modern replication of the original 18th-century wall covering.

and explorer **Charles Marie de La Condamine** (1701–74), went to Peru and Ecuador. When the teams reported their findings, they proved that Newton, not Cassini, was right—**Earth is fatter at the equator**. Also in

1735, as the French exploratory teams set sail, English meteorologist George Hadley (1685–1768) had a key insight into the **trade winds** that drive ships across the Atlantic: **these winds blow east-west, not straight toward the equator** because they are deflected by Earth's rotation.

This year, too, English clockmaker **John Harrison** (1693–1776) completed the first version of his **marine chronometer**, a clock that could keep time accurately enough at sea to allow **longitude to be calculated**. By 1759, Harrison

Linnaeus's animal kingdom

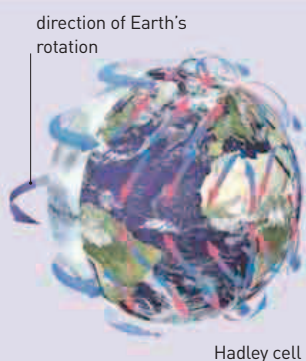
In this table from Systema Naturae, Carl Linnaeus sets out his six subdivisions of the animal kingdom: mammals, birds, amphibians, fish, insects, and worms.

had produced a fourth, pocket-sized version known as H4 that was of even greater accuracy (see 1759–64).

Three years before Maupertius and his team went to Lapland, Swedish naturalist **Carl Linnaeus** had traveled there to collect plant and bird specimens. It was this trip that planted the seeds for his great scheme for **classifying life**, the *Systema Naturae* (*System of Nature*), first published in 1735. **Linnaeus divided the natural world** into three kingdoms—animal, plant, and mineral—and **subdivided each into class, order, genus, and species**. He introduced the now internationally recognized Latin **binomial** (two-part name) classification system which indicates first the genus and then the species.

HADLEY CELL

It is now known that there are three great bands of vertical air circulation, or “cells,” on either side of the equator, including the tropical cell named after the meteorologist George Hadley. The east-west deflection of these cells caused by Earth’s rotation creates a corkscrew pattern that accounts for prevailing winds at different latitudes.



LATE IN THE EVENING OF MAY 28, 1737, English physician and astronomer **John Bevis** (1695–1771) witnessed a rare event through a telescope at the Royal Greenwich Observatory in London: a **planetary occultation**—in which one celestial body passes in front of another, temporarily hiding it from view. What Bevis watched was Venus passing in front of Mercury, the only planetary occultation ever recorded.

In Switzerland, mathematician **Daniel Bernoulli** (1700–82) published *Hydrodynamics*, a **study of the flow of water**, based on his work in St. Petersburg, Russia. Bernoulli noted that as the speed of moving fluid increases, the pressure within it decreases—a phenomenon now known as **Bernoulli's principle**. Also in St. Petersburg, **French astronomer Joseph-Nicolas Delisle** (1688–1768) established a method for **tracking sunspots** as they moved across the Sun.

In 1739, French explorer **Jean-Baptiste Charles Bouvet de Lozier** (1705–86) found the **world's most remote island**, now called Bouvet Island, in the South Atlantic Ocean. In France, physicist **Émilie du Châtelet** (1706–49) published her 1739 paper on combustion, in which she predicted the existence of



CARL LINNAEUS
(1707–78)

Born in Rashult, Sweden, Carl Linnaeus was one of the greatest naturalists of his time. A practicing physician, he spent most of his time classifying plants. His students traveled the world, sending back samples and spreading Linnaean theories. He became Professor of Botany at the University of Uppsala in 1741.

what is now recognized as infrared radiation. In Anjou, France, young Scottish philosopher **David Hume** completed his ***A Treatise on Human Nature***, in which he tried to create a complete psychological profile of man.

1735 English clockmaker John Harrison completes his **first marine chronometer**

1736 Charles Marie de la Condamine discovers rubber

May 28, 1737 John Bevis observes the occultation of Venus

1739 Émilie du Châtelet predicts the existence of infrared radiation

January 1, 1739
Jean-Baptiste
Bouvet discovers
an island later
named after him

1739 David Hume completes *A Treatise on Human Error*

1739 Carl Linnaeus
publishes *Genera*
Plantarum (Types
of Plants)